PES-VCS

os orange level problem
name : Shreecharana
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SCREENSHOTS

1A.

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ make test_objects
gcc -o test_objects test_objects.o object.o -lcrypto
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B.

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

2A.

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B.

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ find .pes/objects -type f
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/f8/0b7bd19a266bddac1ef212fc8cf730d6a4494628e8a182e7c506e07f6c6cac
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/f0/7cb5e083112daea9c88145dc1bd079d051bc3a81530eefec49cb8349b147b0
.pes/objects/c9/a2f1f7227f001a93eaf576e6df63ee6b8a1455b6cfc7aeff6ac812e6fe1748
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ xxd .pes/objects/XX/YYY... | head -20
xxd: .pes/objects/XX/YYY...: No such file or directory
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ xxd .pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39 | head -20
00000000: 626c 6f62 2036 0048 656c 6c6f 0a         blob 6.Hello.
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$
```

3A.

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ make pes
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ echo "hello" > file1.txt
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ echo "world" > file2.txt
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ ./pes add file1.txt file2.txt
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_objects.c
  untracked:  object.c
  untracked:  test_tree.c
  untracked:  README.md
  untracked:  .gitignore
  untracked:  tree.c
  untracked:  test_tree
  untracked:  index.h
  untracked:  commit.c
  untracked:  pes.h
  untracked:  pes.c
  untracked:  tree.h
  untracked:  index.c
  untracked:  Makefile
  untracked:  test_objects
  untracked:  commit.h
  untracked:  test_sequence.sh
```

3B.

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776791342 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776791349 6 file2.txt
```

4A.

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ ./pes log
commit f07cb5e083112daea9c88145dc1bd079d051bc3a81530eefec49cb8349b147b0
Author: PES User <pes@localhost>
Date:   1776791435

    Add farewell

commit f80b7bd19a266bddac1ef212fc8cf730d6a4494628e8a182e7c506e07f6c6cac
Author: PES User <pes@localhost>
Date:   1776791418

    Add world

commit c9a2f1f7227f001a93eaf576e6df63ee6b8a1455b6cfc7aeff6ac812e6fe1748
Author: PES User <pes@localhost>
Date:   1776791401

    Initial commit
```

4B.

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/c9/a2f1f7227f001a93eaf576e6df63ee6b8a1455b6cfc7aeff6ac812e6fe1748
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f0/7cb5e083112daea9c88145dc1bd079d051bc3a81530eefec49cb8349b147b0
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/objects/f8/0b7bd19a266bddac1ef212fc8cf730d6a4494628e8a182e7c506e07f6c6cac
.pes/refs/heads/main
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$
```

4C.

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ cat .pes/refs/heads/main
f07cb5e083112daea9c88145dc1bd079d051bc3a81530eefec49cb8349b147b0
```

FINAL

```
(base) shreecharana@Shreecharana:~/PES1UG24CS917-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: b38fd1edcf85... Initial commit

Log after first commit:
commit b38fd1edcf8589fdf593bba4090190c285582e7e8679861536728ec85e80fbba
Author: PES User <pes@localhost>
Date:   1776791457

    Initial commit


--- Second Commit ---
Committed: 7268a7b6e371... Update file.txt

--- Third Commit ---
Committed: 070b129b7044... Add farewell

--- Full History ---
commit 070b129b704436859bca5fcfc4110462bdc8892169cb0812d8f4c696f249afd4
Author: PES User <pes@localhost>
Date:   1776791458

    Add farewell

commit 7268a7b6e371e00d03f434f1913754feaf3e753021baf7e063af1c32715f5893
Author: PES User <pes@localhost>
Date:   1776791458

    Update file.txt
```

```
commit b38fd1edcf8589fdf593bba4090190c285582e7e8679861536728ec85e80fbba
Author: PES User <pes@localhost>
Date:   1776791457

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
070b129b704436859bca5fcfc4110462bdc8892169cb0812d8f4c696f249afd4

--- Object Store ---
Objects created:
10
.pes/objects/07/0b129b704436859bca5fcfc4110462bdc8892169cb0812d8f4c696f249afd4
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/72/68a7b6e371e00d03f434f1913754feaf3e753021baf7e063af1c32715f5893
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/b3/8fd1edcf8589fdf593bba4090190c285582e7e8679861536728ec85e80fbba
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

Questions

Q5.1: Checkout Implementation

To implement `pes checkout <branch>`, the system first reads the commit hash stored in `.pes/refs/heads/<branch>` and updates `.pes/HEAD` to point to that branch. It then loads the corresponding commit object and reconstructs the working directory from its tree structure by creating, modifying, or deleting files as needed. The index is also updated to match the checked-out tree to maintain consistency. This operation is

complex because it involves modifying multiple files and directories. It must also preserve file permissions and avoid overwriting uncommitted changes. Handling these safely makes checkout a non-trivial process.

Q5.2: Dirty Working Directory Detection

To detect a dirty working directory, the system compares the working directory with the index to identify locally modified files. It then compares the current branch's version of these files with the target branch's version. If a file has local changes and also differs between the two branches, a conflict occurs. In such cases, the checkout operation must be aborted to prevent data loss. This ensures that user modifications are not accidentally overwritten. The index serves as a reliable reference for detecting these changes.

Q5.3: Detached HEAD

In a detached HEAD state, the HEAD points directly to a commit hash instead of a branch reference. New commits can still be created, but they are not linked to any branch. As a result, these commits may become unreachable if the user switches to another branch. This makes them vulnerable to being lost during garbage collection. To

recover such commits, the user can create a new branch pointing to that commit. Alternatively, the commit hash can be retrieved from logs and reused.

Phase 6: Garbage Collection and Space Reclamation

Q6.1: Garbage Collection Algorithm

Garbage collection is implemented using a mark-and-sweep algorithm. First, all reachable objects are marked starting from branch references and HEAD. This includes commits, their associated trees, and all related blobs. A hash set is used to efficiently track visited objects. After marking, any unmarked objects in `.pes/objects` are considered unreachable and are deleted. This process helps reclaim storage space. Even in large repositories, shared history reduces the number of objects that need to be traversed.

Q6.2: Race Condition in Garbage Collection

Running garbage collection concurrently with a commit operation can cause race conditions. During a commit, objects such as blobs and trees are created before

updating the branch reference. If garbage collection runs at this point, it may incorrectly identify these objects as unreachable and delete them. This results in a corrupted commit with missing data. To prevent this, systems use locking mechanisms and delay deletion of recent objects. Real implementations also use reflogs to protect newly created commits.